

Least Squares in Quarto

Your Name

Model

We assume a linear relationship $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. In matrix form, the least-squares estimator is

$$\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y} \quad (1)$$

Simulation

We simulate data and apply Equation 1 directly.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n = 100
x = rng.uniform(0, 10, n)
y = 2.0 + 0.5 * x + rng.normal(0, 1, n)

X = np.column_stack([np.ones(n), x])
beta = np.linalg.solve(X.T @ X, X.T @ y)

xs = np.linspace(0, 10, 100)
plt.scatter(x, y, alpha=0.6, label="data")
plt.plot(xs, beta[0] + beta[1] * xs, color="red", label="fit")
plt.legend()
plt.show()
```

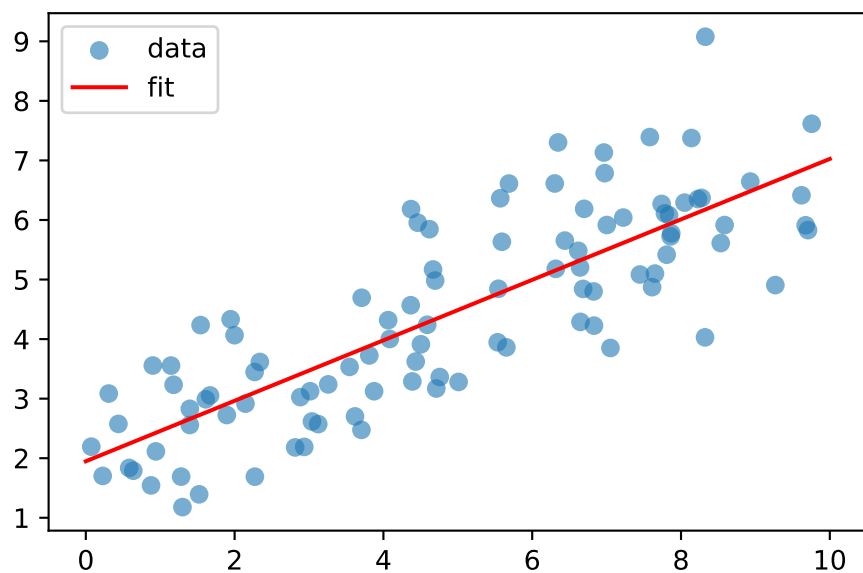


Figure 1: Simulated data and the fitted least-squares line.

Results

As shown in Figure 1, the estimates are $\hat{\beta}_0 = 1.95$ and $\hat{\beta}_1 = 0.51$, close to the true values 2.0 and 0.5.

The residual variance is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

sigma^2 estimate: 0.978